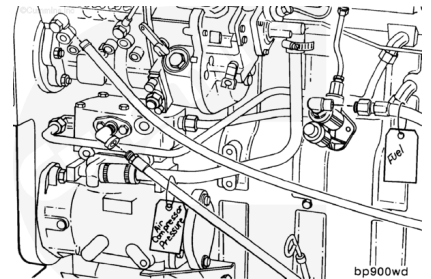


Trainings ▾

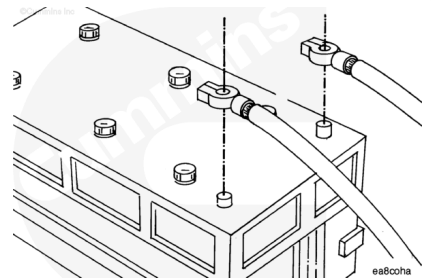
General Information

Place a tag on all hoses, lines, linkages, and electrical connections, as they are removed, to identify their locations on the engine.



⚠ WARNING ⚠

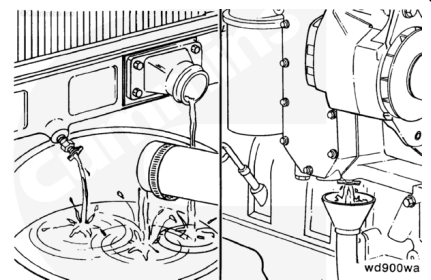
Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the battery cables.

⚠ WARNING ⚠

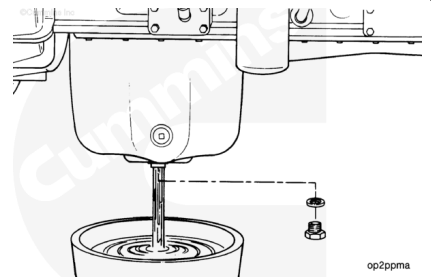
Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.



Drain the engine coolant; refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).

⚠ WARNING ⚠

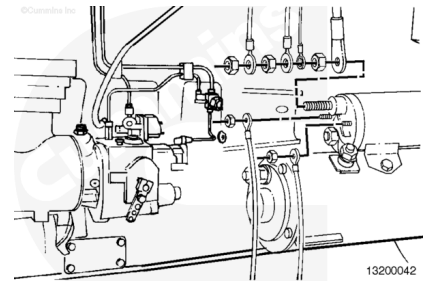
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.



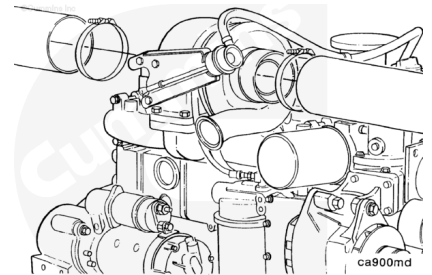
Drain the lubricating oil; refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

Note : Dispose of used oil in accordance with federal, state, and local laws and regulations.

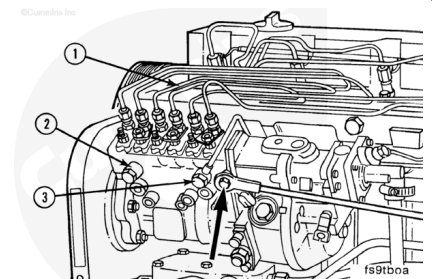
Disconnect the starting motor cable, the engine ground straps, the cab or chassis to engine hoses, the tubing, the electrical wires, and the hydraulic lines.



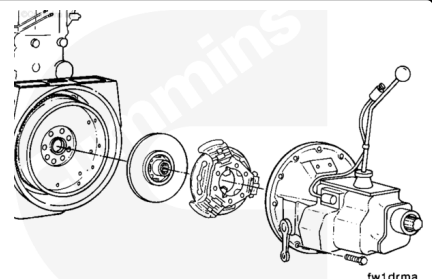
Disconnect the intake and the exhaust air pipes.
Disconnect all chassis-mounted "engine-driven" accessories.



Disconnect the throttle linkage from the injection pump control lever; refer to the OEM service manual.

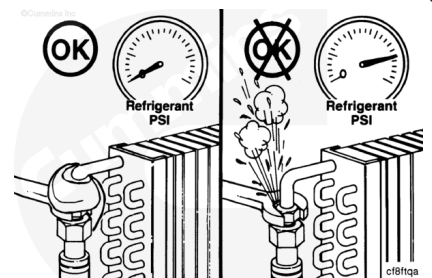


Disconnect the drive units from the flywheel.



⚠ WARNING ⚠

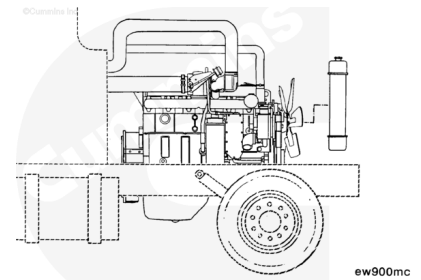
If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fittings before removing. Liquid refrigerant can cause serious eye and skin injuries.



⚠ CAUTION ⚠

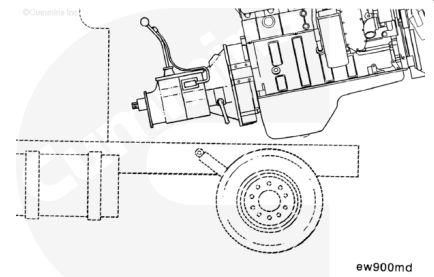
To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

Remove all chassis components necessary to remove the engine from the equipment. Refer to the OEM's instructions.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this assembly.



The engine lifting equipment **must** be designed to lift the engine and transmission safely as an assembly.

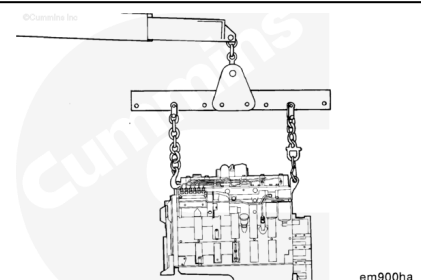
Note : On applications where the rear engine mounts are attached to the transmission, it will probably be necessary to remove the engine and transmission as an assembly.

Note : Use a properly rated hoist and engine lifting fixture.

Note : If the transmission is **not** removed, place a support under the transmission to prevent it from falling.

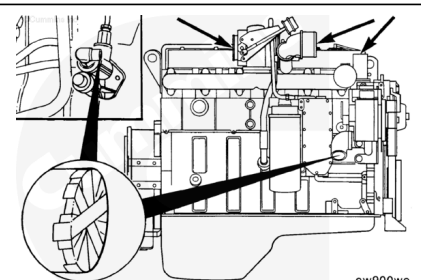
Attach the engine lift fixture, Part No. 3163264, to the engine-mounted lifting brackets.

Remove the engine.

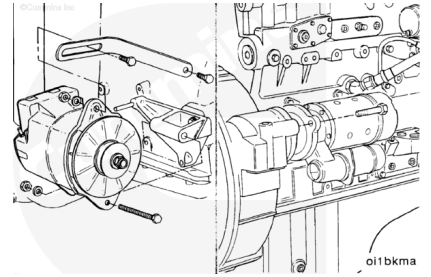


Cover all engine openings to prevent dirt and debris from entering the engine.

Place the engine on suitable engine support stands.



Remove all of the remaining accessories and brackets to use with the replacement engine.



Last Modified: 07-Dec-2004
